

EXP -5

EDA – DATA VISUALIZATION

AIM

To explore and understand the underlying patterns, distributions, trends, and relationships in the data using different visualization techniques such as line charts, bar charts, and histograms.

PROGRAM

```
import matplotlib.pyplot as plt
```

```
import pandas as pd
```

```
import numpy as np
```

```
data = {  
    'Category': ['A', 'B', 'C', 'D'],  
    'Values': [3, 7, 5, 4]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

```
x = np.linspace(0, 10, 100)
```

```
y = np.sin(x)
```

```
plt.plot(x, y, label='Sine Wave')
```

```
plt.xlabel('X-axis')
```

```
plt.ylabel('Y-axis')
```

```
plt.title('Line Plot')
```

```
plt.legend()
```

```
plt.show()
```

```
categories = ['A', 'B', 'C', 'D']
```

```
values = [3, 7, 5, 4]
```

```
plt.bar(categories, values)
```

```
plt.xlabel('Category')
```

```
plt.ylabel('Values')
```

```
plt.title('Bar Plot')
```

```
plt.show()
```

```
data = np.random.randn(1000)
```

```
plt.hist(data, bins=20, edgecolor='black')
```

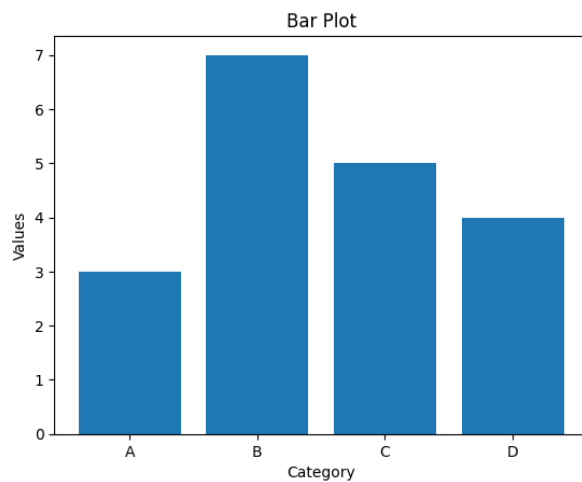
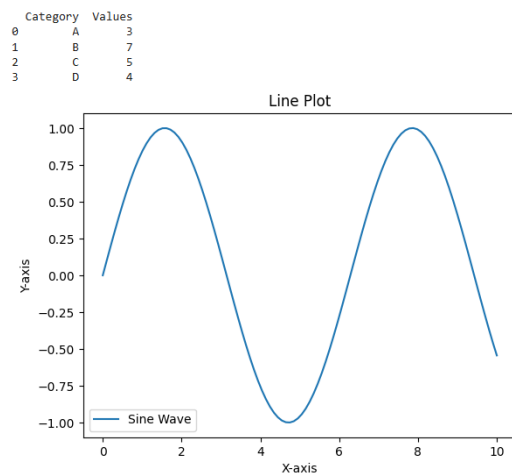
```
plt.xlabel('Value')
```

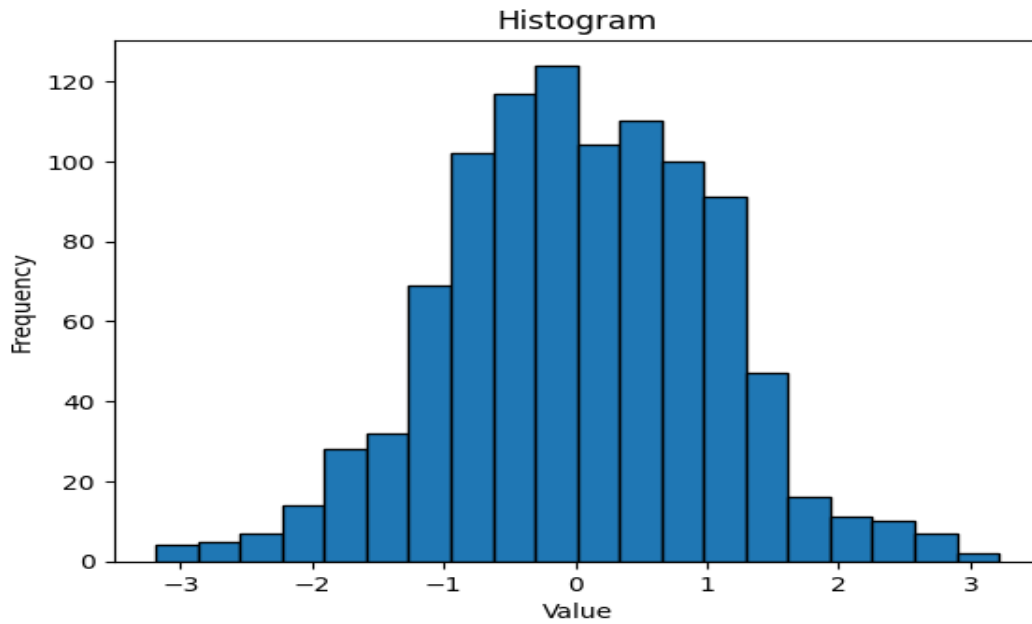
```
plt.ylabel('Frequency')
```

```
plt.title('Histogram')
```

```
plt.show()
```

OUTPUT





RESULT

Thus, the given program was successfully executed. Line chart, bar chart, and histogram were generated successfully to visualize trends, category-wise comparisons, and frequency distributions in the data.